

Data Assimilation in Nonlinear Fluid Flow I

The Kalman filter: exact linear theory, fluid-flow estimation, and nonlinear model mismatch

Presentation 1 of 4

1. Learning objectives

- Formulate fluid-flow estimation as a state-space problem.
- Derive the classical Kalman filter for linear–Gaussian dynamics.
- Build a synthetic nonlinear Burgers-flow experiment.
- Diagnose why a plain KF fails when its linear model cannot represent the nonlinear flow map.

Key idea

Data assimilation is not “just denoising”: it combines dynamics, observations, and uncertainty.

2. Why data assimilation in fluid mechanics?

A numerical solver predicts a complete flow field, but reality is only partially observed.

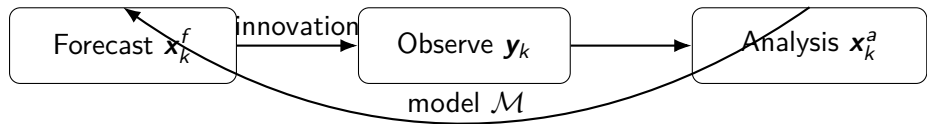
Model provides

- spatially complete state,
- conservation structure,
- temporal continuity,
- a forecast beyond sensors.

Measurements provide

- contact with reality,
- correction of initial-condition error,
- correction of model drift,
- information about unresolved forcing.

3. The forecast–observe–correct loop



$$\text{innovation} = \mathbf{y}_k - \mathcal{H}(\mathbf{x}_k^f).$$

The analysis should move the forecast toward observations *without blindly replacing the model state*.

4. State-space formulation

For a discretized flow field $\mathbf{x}_k \in \mathbb{R}^n$,

$$\mathbf{x}_{k+1} = \mathcal{M}(\mathbf{x}_k) + \mathbf{w}_k, \quad \mathbf{y}_k = \mathcal{H}(\mathbf{x}_k) + \mathbf{v}_k,$$

with model error $\mathbf{w}_k$ and observation error $\mathbf{v}_k$.

Bayesian objective

Compute or approximate

$$p(\mathbf{x}_k \mid \mathbf{y}_{1:k}),$$

the posterior distribution of the current flow state given all measurements so far.

5. Bayesian update in one line

Bayes' rule gives

$$p(\mathbf{x}_k \mid \mathbf{y}_{1:k}) \propto p(\mathbf{y}_k \mid \mathbf{x}_k) p(\mathbf{x}_k \mid \mathbf{y}_{1:k-1}).$$

- $p(\mathbf{x}_k \mid \mathbf{y}_{1:k-1})$: **forecast/prior**, transported through the dynamics.
- $p(\mathbf{y}_k \mid \mathbf{x}_k)$: **likelihood**, determined by sensors and their noise.
- $p(\mathbf{x}_k \mid \mathbf{y}_{1:k})$: **analysis/posterior**.

For nonlinear fluid dynamics, this density is generally non-Gaussian and expensive to represent exactly.

6. Tutorial PDE: viscous Burgers equation

We use the one-dimensional viscous Burgers equation,

$$\frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} = \nu \frac{\partial^2 u}{\partial x^2}, \quad x \in [0, 2\pi),$$

with periodic boundary conditions.

- $u u_x$: nonlinear advection and steepening.
- νu_{xx} : viscous diffusion and smoothing.
- Simple enough to run in a few lines, but nonlinear enough to expose the KF limitation.

7. From PDE to a nonlinear forecast map

After spatial discretization on N grid points,

$$\mathbf{u}^{k+1} = \mathcal{M}_{\Delta t}(\mathbf{u}^k).$$

For Burgers,

$$\mathcal{M}_{\Delta t}(a\mathbf{u} + b\mathbf{v}) \neq a\mathcal{M}_{\Delta t}(\mathbf{u}) + b\mathcal{M}_{\Delta t}(\mathbf{v})$$

because the advective term contains the product $u u_x$.

Important

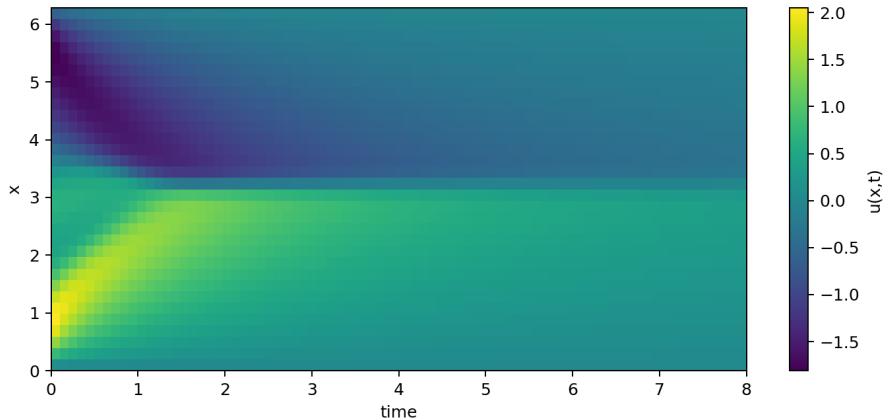
A classical KF requires a linear state transition $\mathbf{x}_{k+1} = \mathbf{F}\mathbf{x}_k + \mathbf{w}_k$.

8. Synthetic experiment setup

Quantity	Value
Grid	$N = 32$ periodic points
Viscosity	$\nu = 0.01$
Assimilation interval	$\Delta t = 0.10$
Number of cycles	80
Observations	every 8th grid point
Observation noise	$\sigma_y = 0.08$
Initial estimate error	Gaussian, std. 0.4

All scripts are in `code/`; the experiment is deterministic because random seeds are fixed.

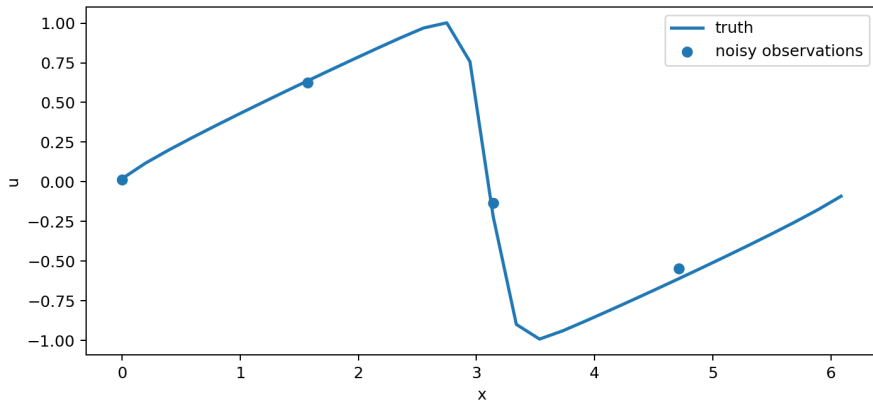
9. Synthetic truth trajectory



Key idea

The state evolves nonlinearly while viscosity gradually damps the field.

10. Sparse noisy observations



- Only 4 of 32 state variables are observed at each cycle.
- The assimilation method must reconstruct unobserved locations through the model/covariance structure.

11. Linear–Gaussian assumptions of the KF

The classical KF assumes

$$\begin{aligned}\mathbf{x}_{k+1} &= \mathbf{F}\mathbf{x}_k + \mathbf{w}_k, & \mathbf{w}_k &\sim \mathcal{N}(0, \mathbf{Q}), \\ \mathbf{y}_k &= \mathbf{H}\mathbf{x}_k + \mathbf{v}_k, & \mathbf{v}_k &\sim \mathcal{N}(0, \mathbf{R}).\end{aligned}$$

If the prior is Gaussian, every forecast and analysis remains Gaussian:

$$\mathbf{x}_k \mid \mathbf{y}_{1:k} \sim \mathcal{N}(\hat{\mathbf{x}}_k, \mathbf{P}_k).$$

The entire posterior is therefore represented by only a mean and covariance.

12. KF forecast equations

Given analysis $(\hat{\mathbf{x}}_{k-1}^a, \mathbf{P}_{k-1}^a)$,

$$\begin{aligned}\hat{\mathbf{x}}_k^f &= \mathbf{F}\hat{\mathbf{x}}_{k-1}^a, \\ \mathbf{P}_k^f &= \mathbf{F}\mathbf{P}_{k-1}^a\mathbf{F}^\top + \mathbf{Q}.\end{aligned}$$

Interpretation:

- the mean is propagated by the model,
- covariance is stretched/rotated by $\mathbf{F}$,
- $\mathbf{Q}$ injects uncertainty due to model error.

13. KF analysis equations

Define the innovation covariance and Kalman gain,

$$\mathbf{S}_k = \mathbf{H}\mathbf{P}_k^f\mathbf{H}^\top + \mathbf{R},$$

$$\mathbf{K}_k = \mathbf{P}_k^f\mathbf{H}^\top\mathbf{S}_k^{-1}.$$

Then

$$\hat{\mathbf{x}}_k^a = \hat{\mathbf{x}}_k^f + \mathbf{K}_k(\mathbf{y}_k - \mathbf{H}\hat{\mathbf{x}}_k^f),$$

$$\mathbf{P}_k^a = (\mathbf{I} - \mathbf{K}_k\mathbf{H})\mathbf{P}_k^f.$$

Key idea

$\mathbf{K}_k$ balances trust in the model forecast against trust in the observations.

14. Theorem: KF is exact for linear–Gaussian systems

Theorem

Assume a Gaussian prior $\mathbf{x} \sim \mathcal{N}(\mathbf{m}, \mathbf{P})$ and linear observation $\mathbf{y} = \mathbf{H}\mathbf{x} + \mathbf{v}$, $\mathbf{v} \sim \mathcal{N}(0, \mathbf{R})$ independent of $\mathbf{x}$. Then $p(\mathbf{x} | \mathbf{y})$ is Gaussian with mean and covariance given by the Kalman analysis equations.

This is stronger than saying the KF is “a good estimator”: under these assumptions it computes the exact Bayesian posterior.

15. Proof sketch: Gaussian conditioning

The joint vector is Gaussian,

$$\begin{bmatrix} \mathbf{x} \\ \mathbf{y} \end{bmatrix} \sim \mathcal{N} \left(\begin{bmatrix} \mathbf{m} \\ \mathbf{Hm} \end{bmatrix}, \begin{bmatrix} \mathbf{P} & \mathbf{PH}^\top \\ \mathbf{HP} & \mathbf{HPH}^\top + \mathbf{R} \end{bmatrix} \right).$$

The conditional mean formula for a partitioned Gaussian gives

$$\mathbb{E}[\mathbf{x} \mid \mathbf{y}] = \mathbf{m} + \mathbf{PH}^\top (\mathbf{HPH}^\top + \mathbf{R})^{-1} (\mathbf{y} - \mathbf{Hm}),$$

and the Schur-complement formula gives the KF covariance. $\square$

16. Why nonlinear dynamics destroy Gaussian closure

Suppose $x \sim \mathcal{N}(0, \sigma^2)$ and propagate through $f(x) = x^2$. Then

$$\mathbb{E}[f(x)] = \mathbb{E}[x^2] = \sigma^2, \quad f(\mathbb{E}[x]) = 0.$$

Also x^2 is not Gaussian.

Important

For nonlinear $\mathcal{M}$, propagating only the mean and covariance is not generally exact; the forecast distribution can become skewed or multimodal.

17. What “plain KF on Burgers” means here

Because the standard KF needs a matrix $\mathbf{F}$, we deliberately use a frozen linear approximation,

$$u_t + c u_x = \nu u_{xx}, \quad c = 0,$$

so the KF forecast contains diffusion but omits nonlinear self-advection.

- Truth: nonlinear Burgers solver.
- KF model: fixed linear diffusion operator.
- Sensors: the sparse noisy observations defined above.

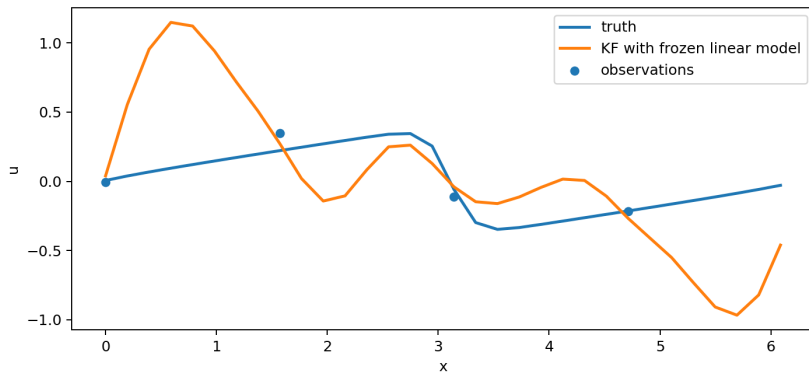
The experiment isolates **model-class mismatch**.

18. Minimal KF code pattern

```
for k in range(1, n_cycles):  
    x = A @ x  
    P = A @ P @ A.T + Q  
  
    S = H @ P @ H.T + R  
    K = solve(S, H @ P).T  
  
    x = x + K @ (y[k] - H @ x)  
    P = (I - K @ H) @ P
```

Full runnable implementation: `code/da_methods.py`.

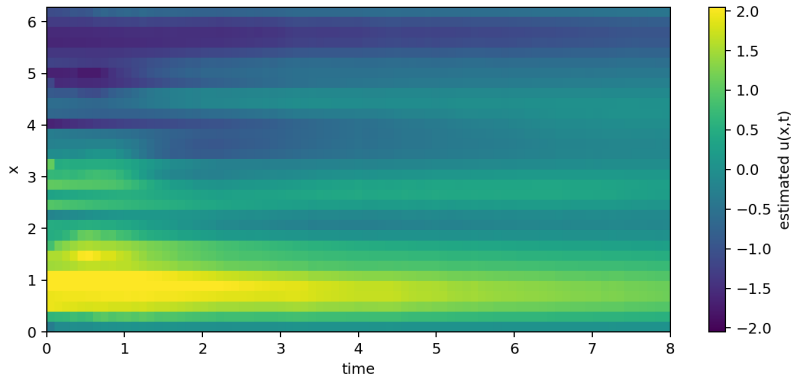
19. KF final-state reconstruction



Key idea

The sparse observations cannot compensate for a forecast model that transports structures incorrectly between sensors.

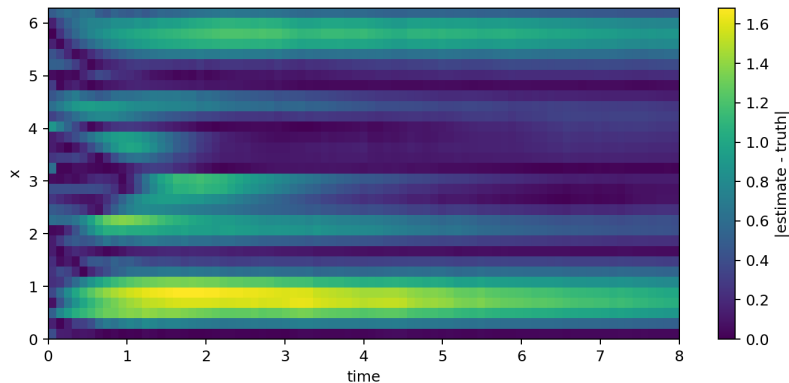
20. KF output over the complete trajectory



Key idea

The KF does not merely miss the final state: the wrong transport pattern persists through much of the assimilation window.

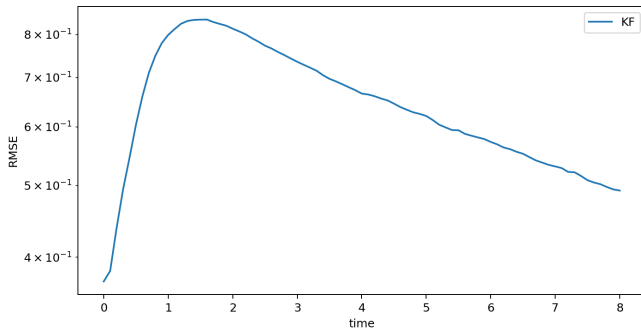
21. KF absolute error map



Important

The colorbar is $|u_{\text{KF}} - u_{\text{truth}}|$. Large coherent regions show systematic model error, not isolated sensor noise.

22. KF RMSE through time



Metric	KF result
Mean trajectory RMSE	0.647
Observed state fraction	$4/32 = 12.5\%$
Nonlinear term represented in forecast	no

23. Why the KF fails here—step by step

- ① Truth transports and steepens structures through $u u_x$.
- ② The KF forecast model only diffuses the field.
- ③ Forecast errors therefore develop a structured spatial bias.
- ④ The covariance P^f is propagated by the same wrong linear dynamics.
- ⑤ Sparse observations correct four points, but the gain spreads those corrections according to the wrong error geometry.

Key idea

A mathematically optimal filter for the *wrong state-space model* can still be a poor estimator of the physical system.

24. Accuracy tradeoff: what the KF assumes versus what the flow does

Component	KF assumption	Burgers-flow reality
Forecast map	fixed linear F	nonlinear $\mathcal{M}_{\Delta t}$
Distribution	Gaussian closure	nonlinear distortion of density
Error transport	FPF^T	state-dependent error directions
Sparse correction	reliable if P^f is reliable	weak when cross-covariance is wrong
Long window	repeated optimal updates	repeated model bias accumulation

25. Computational tradeoff of a full Kalman filter

Object / operation	Storage	Typical dense cost
State $x \in \mathbb{R}^n$	$\mathcal{O}(n)$	$\mathcal{O}(n)$
Covariance $P \in \mathbb{R}^{n \times n}$	$\mathcal{O}(n^2)$	—
Covariance forecast $FPF^\top$	—	up to $\mathcal{O}(n^3)$
Gain construction	—	dense linear algebra

For $n = 10^6$, a dense covariance contains 10^{12} entries. Even before nonlinear-model mismatch, direct full-state KF covariance handling is impractical for large CFD states.

26. When is the classical KF a strong choice?

Good fit	Poor fit
Dynamics accurately linear or linearized over the forecast interval	Strong state-dependent nonlinear transport
State dimension small enough for covariance algebra	Millions of CFD degrees of freedom
Gaussian uncertainty is a reasonable description	Strongly skewed or multimodal uncertainty
Accurate Q and R can be specified	Dominant unresolved / structural model error
Need a rigorous mean/covariance estimate	Need a scalable high-dimensional flow estimator

Key idea

The tradeoff is clear: elegant exact inference under restrictive assumptions versus limited robustness and scaling when those assumptions break.

27. Main takeaways

- Data assimilation combines a forecast model, measurements, and uncertainty into a corrected flow estimate.
- For linear–Gaussian systems, the Kalman analysis is the exact Bayesian posterior mean and covariance.
- Nonlinear Burgers dynamics violate the linear forecast assumption and destroy exact Gaussian closure.
- In the synthetic test, the frozen linear KF gives mean RMSE **0.647**; the error map reveals coherent model-induced errors across space and time.
- Full covariance storage and propagation also become prohibitive as the fluid state dimension grows.